

Short Exam N°2- Computer Vision 2025

1 h 10 – No document allowed (9 questions counting for 15 points)

Part 1: Knowledge questions.

Question 1 (1 point - 5 mn): Must the intrinsic parameters of a camera be estimated again if we move the camera?

Ans: NO, except if we change the distance of the setup along the viewing direction, in such case to avoid blur the focal length may change).

Question 2 (1 point - 5 mn): We use a camera CCD 1/2' which sensor size equals 6.4×4.8 mm. Its resolution equals 800×600 pixels. We use a 25 mm focal lens. Calculate the size of the pixels. Calculate the value of s_x (i.e. the number of pixels per unit of length along direction x). Calculate the value of f_x (that) the calibration must provide.

Ans:

1. square pixels with a size of: $6.4 \text{ mm} / 800 = 4.8 \text{ mm} / 600 = 8 \text{ } \mu\text{m}$.
2. $s_x = s_y = 1/8 \text{ } \mu\text{m} = 125 \text{ mm}^{-1}$
3. $f_x = f_y = f * s_x = 25 \text{ mm} * 125 \text{ mm}^{-1} = 3125$

Question 3 (2 points - 10 mn): We want to estimate a camera model (defined by 12 intrinsic parameters: five intrinsic parameters + seven parameters for a 3rd order radial distortion). Can we calibrate this camera with the photogrammetric approach (multi-view approach) using three views of a 9-point target (and keeping intrinsic parameters constants for each view) ? Explain.

Ans: Yes, the number of unknowns is: $12+6*2+9*3= 51$, meanwhile the number of equations is $9*2*3 = 54$. More unknowns than equations!

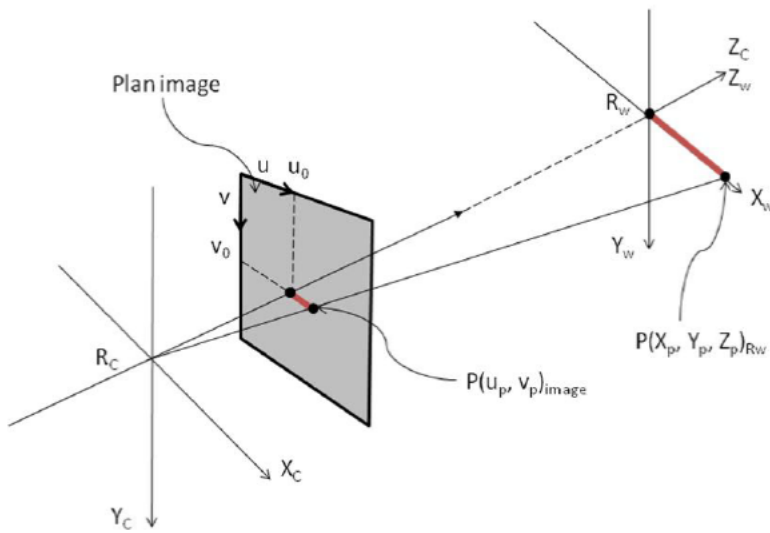
Question 4 (2 points - 5 mn): The calibration of a camera with a resolution of 800× 600 pixels (sensor with square pixels) provided the following results:

$O_x = 225$ and $O_y = 280$ (for the shift of optical center coordinates) and $f_x = 3125$ and $f_y = 4510$.

What enables us to suspect that the calibration is wrong?

Ans: we must find $O_x = 400$ and $O_y = 300$ (i.e. the optical center coordinates should be localized on the center position of the image) and since the pixels are square ones, we must find $f_x = f_y$.

Part 2: Camera calibration.



Question 4 (3 points - 15 mn): Compute the pixel coordinates (u_p, v_p) of the point P considering:

Focal of the camera: $f = 50$ mm

Size of the pixels: $dx = dy = 24$ microns

Length of the red segment in the world coordinate frame (see figure above): 50 cm

Distance of the red segment to the camera: 5 m

Coordinates of the optical center in the image plane: $u_0 = 512, v_0 = 384$

Ans:

$$\begin{bmatrix} su_p \\ sv_p \\ s \end{bmatrix} = \begin{bmatrix} 1/dx & 0 & u_0 \\ 0 & 1/dy & v_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & T_x \\ 0 & 1 & 0 & T_y \\ 0 & 0 & 1 & T_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} R_{11} & R_{12} & R_{13} & 0 \\ R_{21} & R_{22} & R_{23} & 0 \\ R_{31} & R_{32} & R_{33} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X_p \\ Y_p \\ Z_p \\ 1 \end{bmatrix}$$

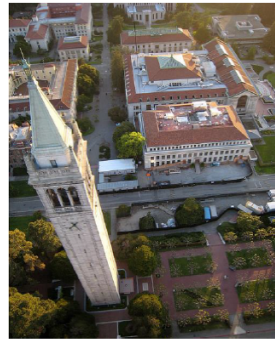
$$\begin{bmatrix} su_p \\ sv_p \\ s \end{bmatrix} = \begin{bmatrix} 1/24e-6 & 0 & 512 \\ 0 & 1/24e-6 & 384 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 50e-3 & 0 & 0 & 0 \\ 0 & 50e-3 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0.5 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} su_p \\ sv_p \\ s \end{bmatrix} = \begin{bmatrix} 3601 \\ 1920 \\ 5 \end{bmatrix}$$

$$u_p = 720$$

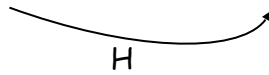
$$v_p = 384$$

Question 5 (2 points - 10 mn): What is the minimal number of point-to-point correspondences to compute (i) homography and (ii) 2D affine geometric transformation? Justify your answer.



Left Image

Right Image



Possible answers:

- a. 1 correspondence for affine transformation, 2 correspondences for homography.
- b. 2 correspondences for affine transformation, 3 correspondences for homography.
- c. 3 correspondences for affine transformation, 4 correspondences for homography.
- d. 4 correspondences for affine transformation, 5 correspondences for homography.

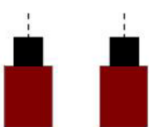
$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & t_x \\ a_{21} & a_{22} & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Affine transformation: 6 DOF $\Rightarrow$ 3 correspondences

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & t_x \\ a_{21} & a_{22} & t_y \\ a_{31} & a_{32} & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Homography transformation: 8 DOF $\Rightarrow$ 4 correspondences (assuming that $a_{33}=1$)

Question 6 (2 points - 10 mn): In the previous image the cameras was aligned as a stereo camera system and where at the same distance to the scene (same intrinsic parameters).

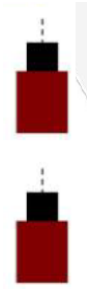


In such situation, how many corresponding points we need to calibrate this stereo setup? And which type of transformation relies the left and right images?

Ans: 6 DOF for the Left camera + 1 DOF (a single translation) for the right camera. With P corresponding points, the number of unknowns is $7 + 3P$ and the number of knowns is $2 \cdot 2P$. Ideally $7 + 3P = 2 \cdot 2P$, i.e. $P=7$.

Ans: A Euclidean (translation) transformation

Question 7 (2 points - 5 mn): Now let us assume that the two cameras are aligned in that way.



In such situation, how many corresponding points we need to calibrate this stereo setup? And which type of transformation relates the left and right images?

Ans: 6 DOF for the nearest camera + 1 DOF (a single translation along the viewing direction) + 1 DOF (for a change of scaling) + 1 DOF (for a possible change of the focal) for the farthest camera. With P corresponding points, the number of unknowns is $9 + 3P$ and the number of knowns is $2 \times 2P$.

Ideally $9 + 3P = 2 \times 2P$, i.e. $P=9$.

Ans: A Similarity (translation+scaling) transformation